Fiscal Year:	FY 2015	**Task Last Updated:**	FY 09/25/2014
PI Name:	Cooper, Maya M.S.		
Project Title:	Food Fortification Stability Study		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Space Human Factors Engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SHFH**:Space Human Factors & Habitability (archival in 2017)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	281.483.1892
Organization Name:	Leidos/NASA Johnson Space Center		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	11/05/2013	**End Date:**	10/30/2014
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmore, Mihriban	**Contact Phone:**	281-244-1004
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 10/30/2014 from 10/01/2016 per PI change per JSC HRP (Ed., 11/1/2016)		
Key Personnel Changes/Previous PI:	PI change in October 2014 to Takiyah Sirmons (previous PI=Maya Cooper). See task with Sirmons as PI for subsequent reporting. Monica Leong began as the PI on this task. Maya Cooper assumed responsibility after Leong left the company and before work was initiated on the task.		
COI Name (Institution):	Douglas, Grace Ph.D. (NASA Johnson Space Center)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NASA is planning for long duration missions beyond low Earth orbit. To ensure sustainment of crew health and performance, adequate nutrition must be delivered by the food system, thereby supporting mission success. The current food system does not meet the shelf life requirements (up to five years) to support long duration missions in part due to the marked degradation of certain vitamins over time. While work continues to determine the magnitude of vitamin loss, a number of vitamins have already been identified that have low concentrations post-processing, or significantly degrade over one to three years of storage. Fortification using commercially available synthetic or encapsulated nutrients, which have increased stability in processing and storage conditions, has been suggested as a countermeasure, but the efficacy and stability of these nutrients is unknown over the extensive storage durations required for exploration missions. It is expected that commercially available fortification nutrients will remain stable through the required shelf life at sufficient levels in compatible foods, processing, and storage conditions. The purpose of this task is to evaluate commercially available fortification nutrients in a selection of space foods under different processing and storage conditions to determine their stability and effect on food acceptability over time. The findings will inform capabilities and limitations of currently available fortification nutrients that will be used to propose requirements for future development.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research, which requires focused and constrained data gathering and analysis that is more appropriately obtained through a non-competitive proposal.
Research Impact/Earth Benefits:	The Food Fortification study will bring vitamin stability knowledge of supplements in food to the open access environment, which will help smaller food manufacturers improve the nutrition of their foods without assuming the prohibitive cost of shelf life research. The data should promote wise fortification of foods. Large-scale companies treat vitamin stability data as proprietary knowledge.
Task Progress:	Five vitamins (vitamin E, vitamin K, pantothenic acid, folic acid, and thiamin) were blended into a vitamin premix (DSM, Freeport, TX) such that the vitamin concentration per serving equaled 25% of the recommended daily intake after two years of ambient storage. Four freeze-dried foods (Scrambled Eggs, Italian Vegetables, Potatoes Au Gratin, Noodles and Chicken) and four thermostabilized foods (Curry Sauce with Vegetables, Chicken Noodle Soup, Grilled Pork Chop, Rice with Butter) were produced, with and without the vitamin premix, to assess the impact of the added fortification on color and taste and to determine the stability of supplemental vitamins in spaceflight foods. The post-processing levels of vitamin indicate that all intended products were fortified but greater amounts of supplement were added to the Grilled Pork Chop and Italian Vegetables than to the other products. Each supplemental vitamin concentration within the Grilled Pork Chop is approximately 2.7 to 3.3 times greater than the end-of-shelf life goal concentration, while the concentrations in the Italian Vegetables range from 2.1 to 3.0 times greater. These two products may prove to have excess fortification at the end of the study. These high nutrient levels could prove a concern for vitamin toxicity, especially in a case where several products were fortified above target levels. The data indicate the necessity of a method for precise control of addition of fortification nutrients in the event that fortification is instituted in the spaceflight food system. In most cases, the fortified and the control products did not differ significantly in overall acceptability when evaluated by the Johnson Space Center volunteer panelists. However, there was a significant difference in the overall acceptability ratings of the Noodles and Chicken fortified and control batches. The panelists' comments on the two Noodles and Chicken products were not significantly different though anecdotally some panelists relayed that there was a higher proportion of chicken in the second sample and chicken hardness may have contributed to lower scores. Comments did not suggest a relationship between acceptability and premix addition. The addition of fortification to spaceflight foods appears to be a plausible mitigation step to inadequate nutrition concerns when considering the ease of adding vitamin premix to the formulations for provisions and the sustainability of the vitamins through the initial processing steps. Vitamin fortification with this premix did not have immediate impact to sensory and color of the food. At this stage, the caveat to introduction is the preciseness to which vitamins can be added. The total amount of vitamin premix required for the largest space production run equates to 10 - 20 grams, a tiny percentage of the formulation. As demonstrated by Italian Vegetables and Grilled Pork Chop, the homogeneity may be difficult to achieve with such small amounts so pouch-to-pouch variability, over-fortification, and under-fortification may ensue. Future analysis should reveal the extent to which this issue should be a concern. PI change in October 2014 to Takiyah Sirmons (previous PI=Maya Cooper). See task with Sirmons as PI for subsequent reporting.
Bibliography Type:	Description: (Last Updated: 04/23/2019)